Evaluation of the Performance of Neural Network-Based Surrogate Models Applied to Multi-Objective Optimization Problems in Bioinformatics

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Abstract. This work addresses the optimization of tryptophan production in Escherichia coli as a case study for a surrogate-assisted multi-objective optimization framework. Tryptophan is an essential amino acid with significant industrial relevance, and its microbial production through E. coli fermentation represents a well-established problem in metabolic engineering.
The problem is formulated with four simultaneous objectives: maximizing cell growth and tryptophan production while minimizing both the glucose supply rate and the number of genetic knockouts applied. Solutions are evaluated using Flux Balance Analysis (FBA) applied to the genome-scale metabolic model iML1515, which is computationally expensive when repeatedly called within an evolutionary loop, motivating the use of surrogate models to approximate the fitness function at a lower cost. A dataset of 4,916 FBA simulations was generated, of which 60.7% resulted in metabolically viable organisms, with viability decreasing progressively as the number of knockouts increased.
NSGA-II was selected as the evolutionary algorithm across all experiments, allowing comparisons to isolate the effect of each surrogate on optimization performance. The models to be implemented and compared include Artificial Neural Networks as the primary model, alongside Random Forest, Support Vector Regression and Kriging as baselines. The resulting Pareto solutions will be validated against real FBA evaluations to quantify approximation error near the optimal region. Problem formulation, dataset creation and initial MOEA setup have been completed, while surrogate model implementation and performance evaluation are still ongoing.

Keywords: Multi-objective optimization, Surrogate models, Tryptophan

1 Introduction

In many real-world applications, particularly in bioinformatics, the evaluation of candidate solutions relies on computationally expensive simulation models, making the optimization process increasingly costly when combined with Multi-Objective

Evolutionary Algorithms (MOEAs), as the iterative nature of these algorithms demands repeated fitness evaluations over many generations (Song et al., 2025).

In this work, the optimization of tryptophan production in *Escherichia coli* is selected as a case study. This problem involves multiple conflicting objectives, maximizing cell growth and amino acid production while minimizing resource consumption, and relies on Flux Balance Analysis (FBA) for solution evaluation, a simulation method that requires solving a linear programming problem at each call. Given the high number of evaluations demanded by evolutionary optimization, the repeated use of FBA becomes computationally prohibitive, making this an ideal and practically relevant scenario for the application of surrogate-assisted multi-objective optimization.

Multi-objective optimization problems involve the simultaneous optimization of conflicting objectives, leading to a set of trade-off solutions known as the Pareto front (Coello et al., 2007; Deb, 2001; Hu et al., 2017). This computational burden represents a major bottleneck in practice, and motivates the use of surrogate models as a more efficient alternative.

Surrogate models, also known as meta-models, are trained to approximate the original evaluation function, providing faster predictions while maintaining acceptable accuracy (Wang et al., 2026). In the context of multi-objective optimization, these models are commonly integrated into MOEAs to replace or reduce calls to the true fitness function, forming what is known as surrogate-assisted optimization frameworks. However, ensuring high surrogate accuracy near the Pareto front remains a critical challenge, as this region is often underrepresented in training data, leading to degraded predictions precisely where they matter most (Díaz-Manríquez et al., 2016).

Therefore, the main objective of this work is to develop an ANN-based surrogate model for multi-objective optimization problems. The specific objectives are:

i. Formulate the optimization of tryptophan production in Escherichia coli as a multi-objective optimization problem ;
ii. Identify the most suitable meta-models (surrogates) for this type of problem;
iii. Implement and adapt these meta-models to the chosen problem;
iv. Integrate the meta-models with multi-objective optimization algorithms (MOEAs);
v. Compare the performance of the meta-models, particularly in the regions close to the Pareto front.

2 State of the art

2.1 Multi-objective Optimization and Surrogates

Multi-objective optimization (MOO) refers to problems involving multiple objective functions to be minimized or maximized simultaneously, subject to constraints, where no single solution optimizes all objectives at once (Coello et al., 2007; Deb, 2001). In contrast to single-objective optimization, MOO considers a multidimensional space where algorithms must determine the limits of the Pareto frontier and identify solutions that lie on it (Deb, 2001). A solution belongs to the Pareto-optimal set when it is impossible to improve any of its objectives without deteriorating at least one other, and the mapping of this set onto the objective space defines the Pareto front (Coello et al., 2007). The main goals in MOO are preserving non-dominated points, maintaining diversity along the Pareto frontier, and providing a representative set of solutions for decision making. The quality of a Pareto front approximation is assessed using performance indicators grouped into cardinality, convergence, distribution and spread (Audet et al., 2021).

Multi-Objective Evolutionary Algorithms (MOEAs have emerged as a prominent approach for addressing problems with multiple conflicting objectives, owing to their inherent ability to simultaneously approximate a diverse set of trade-off solutions across the Pareto front (Hu et al., 2017). These algorithms explore different regions of the solution space and approximate the Pareto front, facilitating decision-making in complex problems (Zhou et al., 2011). Among them, NSGA-II, SPEA2 and MOEA/D are the most widely used (Emmerich & Deutz, 2018; Zhou et al., 2011). NSGA-II is particularly relevant due to its elitism strategy, which preserves the best individuals and improves optimization efficiency (Deb, 2001; Zhan et al., 2025).

The use of surrogate models (meta-models) has been increasingly adopted to reduce expensive fitness function evaluations (Díaz-Manríquez et al., 2016). These models approximate the objective function, reducing computational cost and the need for physical experiments. However, a critical challenge is the degradation of surrogate accuracy near the Pareto front, due to insufficient representation of extreme regions in the training data (Díaz-Manríquez et al., 2016). This limitation may reduce the effectiveness of the evolutionary search as it approaches optimal solutions. Recent studies aim to improve surrogate-assisted evolutionary algorithms (SAEAs) by incorporating local search mechanisms guided by surrogate predictions to better explore promising regions (Li et al., 2024).

Common surrogate models include Kriging, Random Forest (RF), Support Vector Regression (SVR), and Artificial Neural Networks (ANNs). These approaches differ in their modelling strategies. ANNs are a class of machine learning models whose architecture draws inspiration from the organization of neurons in biological nervous systems, composed of multiple layers of artificial neurons that learn highly non-linear relationships directly from data. Their flexibility and representational capacity make them particularly suited to capturing complex non-linear mappings (Wang et al., 2026), which is a desirable property in the context of surrogate-assisted optimization where the relationship between decision variables and objective values can be highly irregular. In this context, Flux Balance Analysis (FBA) is used to evaluate solutions by simulating metabolic flux distributions under stoichiometric constraints (Orth et al., 2010). However, its repeated use requires solving linear programming problems, making it computationally expensive. Therefore, FBA is often integrated with MOEAs and surrogate models to reduce computational cost (Song et al., 2025).

2.2 Tryptophan Production in *E. coli*

Tryptophan is an essential amino acid of significant industrial relevance, finding widespread application across the pharmaceutical, food and animal feed sectors, with a global market exceeding 14,000 tons per year (Ren et al., 2023). Its microbial production through *Escherichia coli* fermentation has emerged as a sustainable and economically viable alternative to chemical synthesis (Niu et al., 2019). The optimization of culture conditions, including glucose supply rate, oxygen availability and genetic modifications such as gene knockouts, with the aim of maximizing tryptophan production while maintaining cell viability, constitutes a well-established problem in metabolic engineering. This framework makes it a suitable and relevant case study for the surrogate-assisted multi-objective optimization framework proposed in this work.

The optimization of tryptophan production is typically supported by the Flux Balance Analysis (FBA) method, described in the previous section, which allows the distribution of intracellular metabolic fluxes under steady-state conditions to be predicted. This approach makes it possible to evaluate how different environmental and genetic perturbations influence metabolic pathways and production levels.

In particular, the analysis of genes involved in the tryptophan biosynthetic pathway, as well as competing metabolic pathways, is fundamental, since genetic modifications such as gene deletions can significantly alter metabolic fluxes. These alterations may enhance tryptophan production by redirecting fluxes towards the desired pathway, or, in some cases, completely inhibit its production. Thus, the systematic exploration of these genetic and environmental factors is essential to identify optimal strategies that maximize tryptophan yield.

3 Work Plan

The project is organized into seven sequential stages, as outlined in **Table 1**, spanning from problem formulation and dataset generation to surrogate model implementation and performance evaluation. At the current stage of development, the first three phases have been successfully completed, while the remaining four are still ongoing.

Table 1. Project overview.

#	Description	Status
1	Find a problem in the field of bioinformatics	Done
2	Formulate it as a Multi Objective Problem (MOP)	Done
3	Create a dataset (using FBA)	Done
4	Run the optimization using MOEA	On progress
5	Implement the meta-models	On progress
6	Validate the Pareto solutions with real FBA	On progress
7	Evaluate and compare the performance of the meta-models	On progress

4 Preliminary Analysis

The chosen problem consists of optimizing the culture conditions of *Escherichia coli* to simultaneously maximize cell growth and tryptophan production while minimizing the glucose supply rate to the reactor and minimizing number of gene knockouts. By reducing the amount of glucose made available to the culture, the cell's actual glucose consumption is indirectly decreased, leading to lower operational costs and improved process efficiency. Tryptophan is an essential amino acid with significant industrial relevance in the pharmaceutical, food and chemical industries, making its microbial production an optimization problem of practical interest.

The evaluation of candidate solutions relies on Flux Balance Analysis (FBA), a widely used approach to simulate cellular metabolism under steady-state conditions that allows the estimation of flux distributions that satisfy biological constraints while optimizing a specific objective function, applied to the genome-scale metabolic model iML1515, which encompasses 1515 genes and over 2700 reactions. Each FBA evaluation requires solving a linear programming problem, making it computationally expensive when repeatedly called within an evolutionary optimization loop. This computational bottleneck motivates the use of surrogate models to approximate the FBA function at a fraction of the cost.

The problem is formulated as a three-objective optimization problem (MOP) with continuous decision variables. The glucose and oxygen uptake rates are sampled uniformly in the range [0.1, 20.0] mmol/gDW/h, producing a continuous and diverse output distribution. The third dimension of the problem is defined by the number of genetic knockouts applied, ranging from 1 to 6 simultaneously silenced genes.

The generated dataset comprises 4,916 FBA simulations, distributed approximately uniformly across the six knockout groups (between 795 and 854 simulations per group). The two output variables considered, biomass growth rate (h^{-1}) and product flux (mmol/gDW/h), exhibit high value diversity. Biomass ranges from 0 to 1.099 h^{-1}, and product flux from 0 to 8.441 mmol/gDW/h, as illustrated in **Fig. 1**. Of the total simulations, 2,982 (60.7%) are metabolically viable, meaning they present positive values for both objectives. The remaining 1,934 simulations (39.3%) resulted in inviable organisms with zero growth, a consequence of knockout and uptake combinations incompatible with cell survival. The viability rate decreases progressively with the number of knockouts, from 85.8% for 1 knockout down to 44.9% for 6 knockouts, reflecting the cumulative impact of genetic perturbation on metabolism.

For the MOEA the NSGA-II algorithm will be considered as the multi-objective optimization algorithm across all experiments. NSGA-II was chosen due to its widespread adoption in the literature for problems with two to three objectives, its elitist selection mechanism grounded in the principles of non-dominated sorting and diversity preservation through crowding distance estimation, and its well-documented performance in surrogate-assisted optimization frameworks. Using a fixed MOEA would allow the comparison to isolate the effect of the surrogate model on optimization performance.

Several surrogate models will be implemented and compared in this project. The primary model is the Artificial Neural Network (ANN), chosen in accordance with the neural network focus defined in the project proposal. To enable a rigorous performance evaluation, additional surrogate models, like Random Forest (RF), Support Vector Regression (SVR), and Kriging, will be used as baselines for comparison. Each model will be independently integrated into the NSGA-II optimization loop as a replacement for the FBA fitness evaluation function, being applied to the candidate solutions generated by the evolutionary algorithm. Performance will be assessed using global accuracy metrics as well as metrics specific to the Pareto-optimal region, where surrogate degradation is expected to be most pronounced. The Pareto solutions produced by each surrogate will be validated against real FBA evaluations to quantify the approximation error near the optimum and determine which model best preserves accuracy in that critical region.

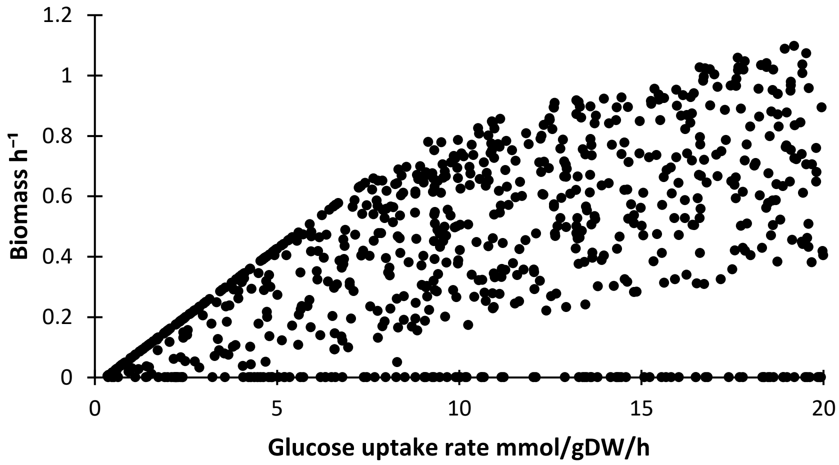

Fig. 1. Example of the distribution of biomass growth rate as a function of glucose uptake rate across all simulations with only one knockout. Points at y = 0 correspond to metabolically inviable solutions.

5 Conclusion

In conclusion, this work addresses a complex and computationally demanding multi-objective optimization problem involving the metabolic engineering of *Escherichia coli* for enhanced tryptophan production. By simultaneously maximizing biomass growth and product formation while minimizing glucose consumption and genetic interventions, the problem reflects realistic industrial constraints and trade-offs.

The preliminary analysis of the dataset indicates that the problem exhibits a considerable level of complexity for both the NSGA-II optimization process and the surrogate modeling approach. The high variability of the outputs, the presence of a significant proportion of infeasible solutions, and the nonlinear relationships between decision variables and objectives contribute to a challenging search space.

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